

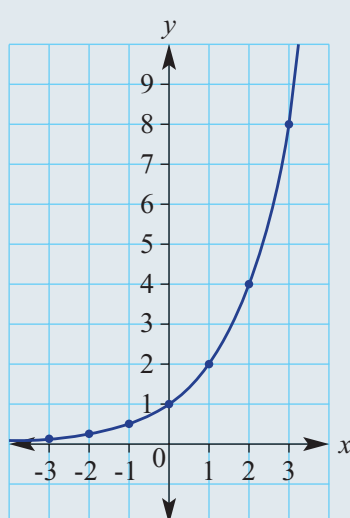
Example 21 Plotting an exponential graphFor the rule $y = 2^x$:**a** complete this table**b** plot points to form its graph

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$				

Solution**Explanation****a**

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

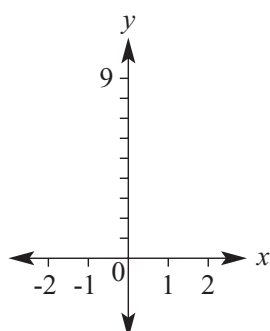
$2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8.$

b

Plot each point and join to form a smooth curve.

7 Consider the exponential rule $y = 3^x$.**a** Complete this table.

x	-2	-1	0	1	2
y	$\frac{1}{9}$	$\frac{1}{3}$			

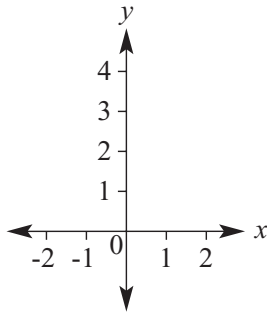
b Plot the points in the table to form the graph of $y = 3^x$.

- 8 Consider the exponential rule $y = 4^x$.

a Complete this table.

x	-2	-1	0	1	2
y	$\frac{1}{16}$	$\frac{1}{4}$			

- b Plot the points in the table to form the graph of $y = 4^x$.



- 9 a Sketch the graph of the following on the same set of axes, labelling the y -intercept and the point where $x = 1$.

i $y = 2^x$

ii $y = 4^x$

iii $y = 5^x$

- b What do you notice about the y -intercept on each graph?
c What does increasing the base number do to each graph?

Problem-solving and Reasoning

Use this table to help.

x	-1	0	1	2
$y = 2^x$	$\frac{1}{2}$	1		
$y = 4^x$	$\frac{1}{4}$			
$y = 5^x$	$\frac{1}{5}$			

- 10 Give the radius of the circles with these equations.

a $x^2 + y^2 = 36$

b $x^2 + y^2 = 81$

c $x^2 + y^2 = 144$

d $x^2 + y^2 = 5$

e $x^2 + y^2 = 14$

f $x^2 + y^2 = 20$

Remember:
 $x^2 + y^2 = r^2$

- 11 Write the equation of a circle with centre $(0, 0)$ and radius 7.

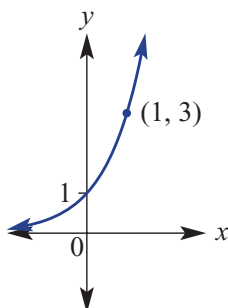
- 12 Match equations a–c with graphs A–C.

a $y = -x - 2$

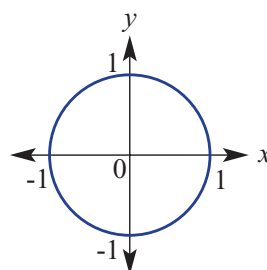
b $y = 3^x$

c $x^2 + y^2 = 1$

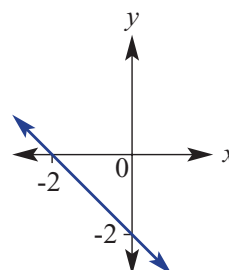
A



B



C



13 A study shows that the population of a town is modelled by the rule $P = 2^t$, where t is in years and P is in thousands of people.

- a** State the number of people in the town at the start of the study ($t = 0$).
b State the number of people in the town after:
i 1 year **ii** 3 years
c When is the town's population expected to reach:
i 4000 people? **ii** 16 000 people?

If $P = 3$, there are 3000 people.



★ Graphs of hyperbolas

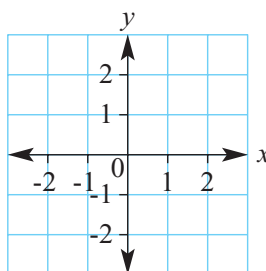
Another type of graph is called a hyperbola, and it comes from the rule $y = \frac{1}{x}$.

14 A hyperbola has the rule $y = \frac{1}{x}$.

- a** Complete this table.

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

- b** Plot the points to form the graph of $y = \frac{1}{x}$.

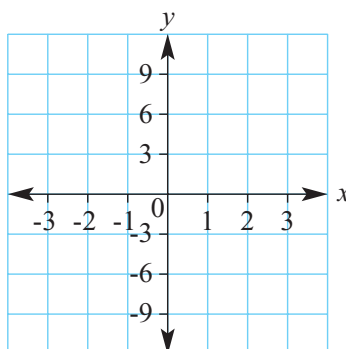


15 A hyperbola has the rule $y = \frac{3}{x}$.

- a** Complete this table

x	-3	-1	$-\frac{1}{3}$	$\frac{1}{3}$	1	3
y						

- b** Plot the points to form the graph of $y = \frac{3}{x}$.



Using technology 10.7: Graphing circles and other graphs

This activity is available on the companion website as a printable PDF.

- 1 I am a beautiful curve! Solve the equations then match the letters to the answers to find out what I am.

O

2^3

B

$\text{radius of } x^2 + y^2 = 16$

A

$\text{solution to } (x - 3)^2 = 0$

L

$x^2 - x$

P

$-2(x - 1)$

R

$\text{solution to } x^2 - 2x - 8 = 0$

$-2x + 2$

3

$4, -2$

3

4

8

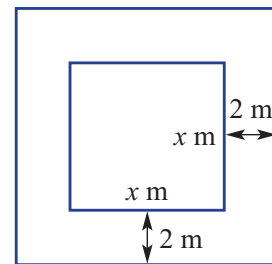
$x(x - 1)$

3

- 2 A square pool of side length x metres is surrounded by a 2 m wide tiled edge.

a Find an expression for the total area.

b For what value of x is the total area equal to 100 m^2 ?



- 3 A book's length is 6 cm longer than its width and its total cover area is 280 cm^2 . What are its dimensions?
- 4 The product of two consecutive even numbers is 168. Find the two numbers.
- 5 A father's age is the square of his son's age (x). In 20 years' time the father will be 3 times as old as his son. What are the ages of the father and son?
- 6 A rectangular painting is to have a total area (including the frame) of 1200 cm^2 . The painting is 30 cm long and 20 cm wide, find the width of the frame.
- 7 Simplify these expressions.
- a $4x - 3(2 - x)$ b $(x - 1)^2 - (x + 1)^2$ c $\frac{x^2 - x - 6}{x + 2}$
- 8 A cyclist in a charity ride rides 300 km at a constant average speed. If the average speed had been 5 km/h faster the ride would have taken 2 hours less. What was the average speed of the cyclist?

